

FILES IN JAVA

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File



A **file** means a **collection of data or information stored in a computer**. It is saved with a **name and extension** so we can open and use it later. 📁

Example

- ❑ document.txt → text file
- ❑ java.pptx → PowerPoint file
- ❑ program.java → Java program file
- ❑ marks.csv → data file

Types of Files

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- ❑ Text file (.txt)
- ❑ Document file (.docx)
- ❑ Presentation file (.pptx)
- ❑ Program file (.java, .py)
- ❑ Data file (.csv)

👉 Example: The Java PPT I sent you is a .pptx presentation file.

Java Introduction

- ❖ Java is a high-level, object-oriented programming language developed by Sun Microsystems in 1995.
- ❖ It is platform independent and widely used for web, mobile, and enterprise applications

Features of Java

Main features of Java:

- ❖ **Platform Independent** - Any OS la run .
- ❖ **Object Oriented** - Use Class and Object .
- ❖ **Secure** - Virus and malware protection .
- ❖ **Robust** - Errors handle panna easy.
- ❖ **Multithreading** - In One time it do many tasks .

Java Architecture

Java architecture main 3 parts:

- ✓ **JDK (Java Development Kit)** - Java program creating tools.
- ✓ **JRE (Java Runtime Environment)** - Program run in environment.
- ✓ **JVM (Java Virtual Machine)** - Bytecode execute .

Flow:

Java Code → Compile → Bytecode → JVM → Output

Java Program Structure

Basic Java program structure:

```
class Hello {  
    public static void main(String[] args){  
        System.out.println("Hello World");  
    }  
}
```

Object Oriented Programming (OOP)

Java follow panna main concepts:

- **Class** – Blueprint
- **Object** – Real entity
- **Inheritance** – Use Parent class properties use
- **Polymorphism** – Same method different work
- **Encapsulation** – Data hide
- **Abstraction** – Only show Important things

Variables and Data Types

Variable – Use to store data

Example data types:

- **int** – Integer numbers
- **float / double** – Decimal numbers
- **char** – Single character
- **boolean** – True / False
- **String** – Text
- Example:
 int a = 10;
 String name = "Niranjana";

Control Statements

Control Program flow.

Types:

- **if / else** - Condition check
- **switch** - Multiple choice condition
- **for loop** - Repeat statements

- while loop – Condition true repeat
- do-while – At least once execute

Classes and Objects

- **Class** → Blueprint
- **Object** → Class instance

Example:

```
class Student {  
    int id;  
    String name;  
}
```

Object create

Program Structure

Basic Java program structure:

```
class Hello {  
    public static void main(String[] args){  
        System.out.println("Hello World");  
    }  
}
```

Parts:

- **Class** - Program container
- **main() method** - Program start point
- **Statements** - Instructions

Java packages

- Packages are used to organize classes and interfaces into groups, helping manage large projects and avoid name conflicts.

Exception handling

- Exception handling manages runtime errors using try, catch, finally, throw, and throws keywords.

Advantages of Java

Write Once Run Anywhere, large community support,
secure environment,
strong memory management

Conclusion

Java remains one of the most popular programming languages used in software development, mobile apps, and enterprise systems.

Write a java program to handle division by zero.

```
class Divide{  
    public static void main(String args[]){  
        try{  
            int a=10,b=0;  
            int c=a/b;  
            System.out.println(c);  
        }  
        catch(Exception e){  
            System.out.println("Cannot divide by zero");  
        }  
    }  
}
```

#output:

Cannot divide by zero